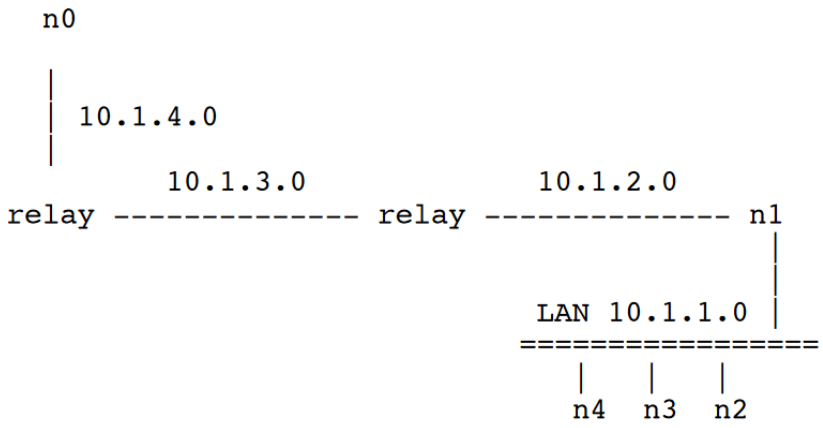
Félix Arnold

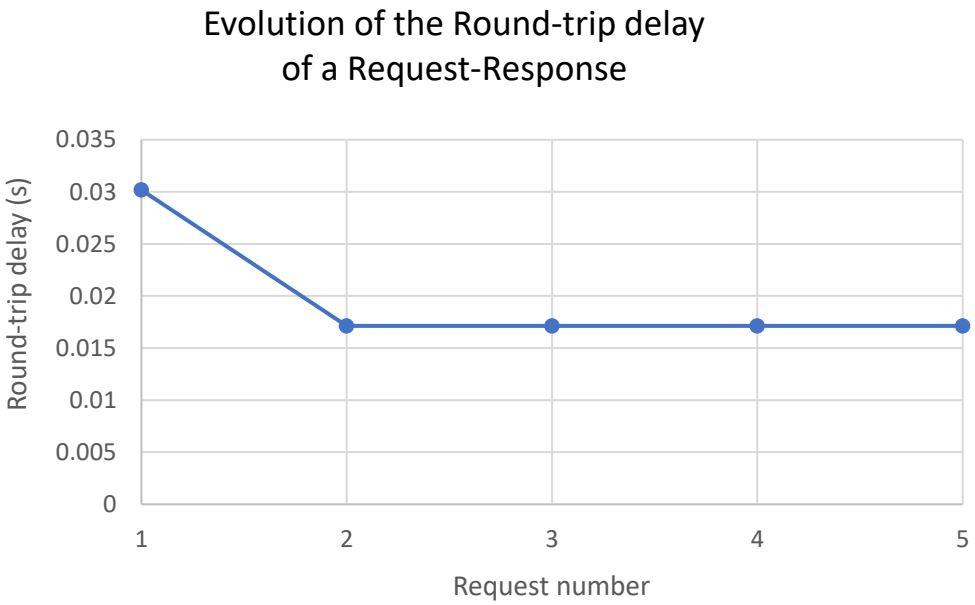
ECE 6110 - Lab Assignment 1
Topologies, Logging, and PCAP Tracing

Part 2

For this last part, we incorporate the Ping application in a modified topology including relay nodes and CSMA nodes. We follow the following example of topology:



For 3 relay nodes, 5 CSMA nodes and 4s of interval between each request, we can plot the following graph of the evolution of round-trip delay.



We observe that the round-trip delay of the first request is significantly more important than the other ones, with a delay increased by 0.013049s. We can find the cause of this in the *pcap* file of the CSMA nodes.

When the CSMA nodes receives the ping request, they must wait for an ARP request to be finished. ARP stands for Address Resolution Protocol and aims at mapping an IP address to a MAC address on a local network. This happens because node n1 knows the IP address to which it must send the ping request, but does not know its MAC address. It therefore broadcasts an ARP request on the CSMA network to obtain it. The destination node, n5 in this example, then returns its MAC address and n1 can forward the ping request. The same thing happens when n5 wants to send the ping packet back to n0. It knows that to reach n0 it needs to send the packet to another node in the CSMA node. Its IP address is known, but not its MAC address. As such, it must broadcast an ARP request to find it.

For the subsequent ping, the IP addresses are already mapped to a MAC address so there is no need to send new ARP requests.

Using the logging in ns3, we can figure out the behavior of the Address Resolution Protocol. Here is a copy of the relevant part of the output of the simulation in the console, with some added comment. The full output, including also the comments, can be found in the attached text file.

```
-----------> The first Ping request is send by the client node.
+0.000000000s Ping:Ping(0x55986e301ab0)
+2.000000000s Ping:StartApplication(0x55986e301ab0)
+2.000000000s Ping:GetApplicationSignature(0x55986e301ab0)
PING 10.1.1.6 - 56 bytes of data; 84 bytes including ICMP and IPv4 headers.

-----------> The node 10.1.1.1 is connected at the same time to the CSMA and point-to-point
network. It receives a Ping packet.
+2.000000000s Ping:Send(0x55986e301ab0)
+2.000000000s Ping:Send(): m_seq=0
+2.000000000s Ping:Write64(0x55986e301ab0, 0x55986e2ca760, 0)

-----------> It checks if it knows a MAC address mapped to the IP address of the Ping packet.
+2.008550400s ArpL3Protocol:Lookup(0x55986e2dd2d0, 0x55986e319300, 10.1.1.6, 0x55986e176f00,
0x55986e1e1690, 0x7ffdf09d2f20)
+2.008550400s ArpL3Protocol:Lookup(): node=4, no entry for 10.1.1.6 -- send arp request

-----------> No MAC address was found, so the node create an ARP request.
+2.017550400s ArpL3Protocol:SendArpRequest(0x55986e2dd2d0, 0x55986e1e1690, 10.1.1.6)
+2.017550400s ArpL3Protocol:SendArpRequest(): ARP: sending request from node 4 || src:
02-06-00:00:00:00:00:09 / 10.1.1.1 || dst: 02-06-ff:ff:ff:ff:ff:ff / 10.1.1.6

-----------> The ARP request is broadcasted to every nodes in the CSMA network. If the IP address
does not match the one of the node, the request is ignored.
+2.017562080s ArpL3Protocol:Receive(0x55986e2e04c0, 0x55986e173910, 46, 2054,
02-06-00:00:00:00:00:09, 02-06-00:00:00:00:00:0a, 0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.017562080s ArpL3Protocol:FindCache(0x55986e2e04c0, 0x55986e173910)
+2.017562080s ArpL3Protocol:Receive(): ARP: received request node=5, got request from 10.1.1.1 for
address 10.1.1.6; we have addresses:
+2.017562080s ArpL3Protocol:Receive(): 10.1.1.2,
+2.017562080s ArpL3Protocol:Receive(): node=5, got request from 10.1.1.1 for unknown address
10.1.1.6 -- drop
+2.017562080s ArpL3Protocol:Receive(0x55986e2e36b0, 0x55986e1736e0, 46, 2054,
02-06-00:00:00:00:00:09, 02-06-00:00:00:00:00:0b, 0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.017562080s ArpL3Protocol:FindCache(0x55986e2e36b0, 0x55986e1736e0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received request node=6, got request from 10.1.1.1 for
address 10.1.1.6; we have addresses:
+2.017562080s ArpL3Protocol:Receive(): 10.1.1.3,
+2.017562080s ArpL3Protocol:Receive(): node=6, got request from 10.1.1.1 for unknown address
10.1.1.6 -- drop
```

```
+2.017562080s ArpL3Protocol:Receive(0x55986e2e6890, 0x55986e2cf250, 46, 2054,
02-06-00:00:00:00:00:09, 02-06-00:00:00:00:00:0c, 0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.017562080s ArpL3Protocol:FindCache(0x55986e2e6890, 0x55986e2cf250)
+2.017562080s ArpL3Protocol:Receive(): ARP: received request node=7, got request from 10.1.1.1 for
address 10.1.1.6; we have addresses:
+2.017562080s ArpL3Protocol:Receive(): 10.1.1.4,
+2.017562080s ArpL3Protocol:Receive(): node=7, got request from 10.1.1.1 for unknown address
10.1.1.6 -- drop
+2.017562080s ArpL3Protocol:Receive(0x55986e2e9d00, 0x55986e2cfeb0, 46, 2054,
02-06-00:00:00:00:00:09, 02-06-00:00:00:00:00:0d, 0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.017562080s ArpL3Protocol:FindCache(0x55986e2e9d00, 0x55986e2cfeb0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received request node=8, got request from 10.1.1.1 for
address 10.1.1.6; we have addresses:
+2.017562080s ArpL3Protocol:Receive(): 10.1.1.5,
+2.017562080s ArpL3Protocol:Receive(): node=8, got request from 10.1.1.1 for unknown address
10.1.1.6 -- drop
+2.017562080s ArpL3Protocol:Receive(0x55986e2ed170, 0x55986e2d0b40, 46, 2054,
02-06-00:00:00:00:00:09, 02-06-00:00:00:00:00:0e, 0)
+2.017562080s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.017562080s ArpL3Protocol:FindCache(0x55986e2ed170, 0x55986e2d0b40)
+2.017562080s ArpL3Protocol:Receive(): ARP: received request node=9, got request from 10.1.1.1 for
address 10.1.1.6; we have addresses:
+2.017562080s ArpL3Protocol:Receive(): 10.1.1.6,

-----------> The node 9 has the same IP address as the one in the ARP request, 10.1.1.6. It sends
back a response.
+2.017562080s ArpL3Protocol:Receive(): node=9, got request from 10.1.1.1 -- send reply
+2.017562080s ArpL3Protocol:SendArpReply(0x55986e2ed170, 0x55986e2fca70, 10.1.1.6, 10.1.1.1,
00-06-00:00:00:00:00:09)
+2.017562080s ArpL3Protocol:SendArpReply(): ARP: sending reply from node 9|| src:
02-06-00:00:00:00:00:0e / 10.1.1.6 || dst: 00-06-00:00:00:00:00:09 / 10.1.1.1

-----------> The node 10.1.1.1 which initiaded the ARP request receives the response with the MAC
address corresponding to the destination IP address of the Ping packet.
+2.017574760s ArpL3Protocol:Receive(0x55986e2dd2d0, 0x55986e176f00, 46, 2054,
02-06-00:00:00:00:00:0e, 02-06-00:00:00:00:00:09, 0)
+2.017574760s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.017574760s ArpL3Protocol:FindCache(0x55986e2dd2d0, 0x55986e176f00)
+2.017574760s ArpL3Protocol:Receive(): ARP: received reply node=4, got reply from 10.1.1.6 for
address 10.1.1.1; we have addresses:
+2.017574760s ArpL3Protocol:Receive(): 10.1.1.1,
+2.017574760s ArpL3Protocol:Receive(): node=4, got reply from 10.1.1.6 for waiting entry -- flush
-----------> Now the destination MAC address is known and the ping packet can be sent to the
destination node of the Ping request (the node 9).
+2.017574760s ArpL3Protocol:Lookup(0x55986e2dd2d0, 0x55986e319300, 10.1.1.6, 0x55986e176f00,
0x55986e1e1690, 0x7ffdf09d3340)
+2.017574760s ArpL3Protocol:Lookup(): node=4, alive entry for 10.1.1.6 valid -- send

-----------> The node 10.1.1.6 needs to respond to the ping request, and need to check if it knows
the MAC address of the previous node.
+2.017590480s ArpL3Protocol:Lookup(0x55986e2ed170, 0x55986e1cfc90, 10.1.1.1, 0x55986e2d0b40,
0x55986e2fca70, 0x7ffdf09d2900)
+2.017590480s ArpL3Protocol:Lookup(): node=9, no entry for 10.1.1.1 -- send arp request

-----------> It does not know it. So as previously, it initiates an ARP request.
+2.021590480s ArpL3Protocol:SendArpRequest(0x55986e2ed170, 0x55986e2fca70, 10.1.1.1)
+2.021590480s ArpL3Protocol:SendArpRequest(): ARP: sending request from node 9 || src:
02-06-00:00:00:00:00:0e / 10.1.1.6 || dst: 02-06-ff:ff:ff:ff:ff:ff / 10.1.1.1

-----------> The ARP request is broadcasted to every nodes in the CSMA network. The node 5
recognizes its IP address, 10.1.1.1 and needs to send a response.
+2.021602160s ArpL3Protocol:Receive(0x55986e2dd2d0, 0x55986e176f00, 46, 2054,
02-06-00:00:00:00:00:0e, 02-06-00:00:00:00:00:09, 0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.021602160s ArpL3Protocol:FindCache(0x55986e2dd2d0, 0x55986e176f00)
+2.021602160s ArpL3Protocol:Receive(): ARP: received request node=4, got request from 10.1.1.6 for
address 10.1.1.1; we have addresses:
+2.021602160s ArpL3Protocol:Receive(): 10.1.1.1,
+2.021602160s ArpL3Protocol:Receive(): node=4, got request from 10.1.1.6 -- send reply
```

```
----------> The other nodes simply ignore the request.
+2.021602160s ArpL3Protocol:SendArpReply(0x55986e2dd2d0, 0x55986e1e1690, 10.1.1.1, 10.1.1.6,
00-06-00:00:00:00:00:0e)
+2.021602160s ArpL3Protocol:SendArpReply(): ARP: sending reply from node 4|| src:
02-06-00:00:00:00:00:09 / 10.1.1.1 || dst: 00-06-00:00:00:00:00:0e / 10.1.1.6
+2.021602160s ArpL3Protocol:Receive(0x55986e2e04c0, 0x55986e173910, 46, 2054,
02-06-00:00:00:00:00:0e, 02-06-00:00:00:00:00:0a, 0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.021602160s ArpL3Protocol:FindCache(0x55986e2e04c0, 0x55986e173910)
+2.021602160s ArpL3Protocol:Receive(): ARP: received request node=5, got request from 10.1.1.6 for
address 10.1.1.1; we have addresses:
+2.021602160s ArpL3Protocol:Receive(): 10.1.1.2,
+2.021602160s ArpL3Protocol:Receive(): node=5, got request from 10.1.1.6 for unknown address
10.1.1.1 -- drop
+2.021602160s ArpL3Protocol:Receive(0x55986e2e36b0, 0x55986e1736e0, 46, 2054,
02-06-00:00:00:00:00:0e, 02-06-00:00:00:00:00:0b, 0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.021602160s ArpL3Protocol:FindCache(0x55986e2e36b0, 0x55986e1736e0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received request node=6, got request from 10.1.1.6 for
address 10.1.1.1; we have addresses:
+2.021602160s ArpL3Protocol:Receive(): 10.1.1.3,
+2.021602160s ArpL3Protocol:Receive(): node=6, got request from 10.1.1.6 for unknown address
10.1.1.1 -- drop
+2.021602160s ArpL3Protocol:Receive(0x55986e2e6890, 0x55986e2cf250, 46, 2054,
02-06-00:00:00:00:00:0e, 02-06-00:00:00:00:00:0c, 0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.021602160s ArpL3Protocol:FindCache(0x55986e2e6890, 0x55986e2cf250)
+2.021602160s ArpL3Protocol:Receive(): ARP: received request node=7, got request from 10.1.1.6 for
address 10.1.1.1; we have addresses:
+2.021602160s ArpL3Protocol:Receive(): 10.1.1.4,
+2.021602160s ArpL3Protocol:Receive(): node=7, got request from 10.1.1.6 for unknown address
10.1.1.1 -- drop
+2.021602160s ArpL3Protocol:Receive(0x55986e2e9d00, 0x55986e2cfeb0, 46, 2054,
02-06-00:00:00:00:00:0e, 02-06-00:00:00:00:00:0d, 0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.021602160s ArpL3Protocol:FindCache(0x55986e2e9d00, 0x55986e2cfeb0)
+2.021602160s ArpL3Protocol:Receive(): ARP: received request node=8, got request from 10.1.1.6 for
address 10.1.1.1; we have addresses:
+2.021602160s ArpL3Protocol:Receive(): 10.1.1.5,
+2.021602160s ArpL3Protocol:Receive(): node=8, got request from 10.1.1.6 for unknown address
10.1.1.1 -- drop
+2.021614840s ArpL3Protocol:Receive(0x55986e2ed170, 0x55986e2d0b40, 46, 2054,
02-06-00:00:00:00:00:09, 02-06-00:00:00:00:00:0e, 0)
+2.021614840s ArpL3Protocol:Receive(): ARP: received packet of size 46
+2.021614840s ArpL3Protocol:FindCache(0x55986e2ed170, 0x55986e2d0b40)
+2.021614840s ArpL3Protocol:Receive(): ARP: received reply node=9, got reply from 10.1.1.1 for
address 10.1.1.6; we have addresses:
+2.021614840s ArpL3Protocol:Receive(): 10.1.1.6,
+2.021614840s ArpL3Protocol:Receive(): node=9, got reply from 10.1.1.1 for waiting entry -- flush

----------> The node 9 now knows the MAC address of the node 10.1.1.1. It can sends its Ping
packet.
+2.021614840s ArpL3Protocol:Lookup(0x55986e2ed170, 0x55986e1cfc90, 10.1.1.1, 0x55986e2d0b40,
0x55986e2fca70, 0x7ffdf09d3340)
+2.021614840s ArpL3Protocol:Lookup(): node=9, alive entry for 10.1.1.1 valid -- send

----------> The client node receives the answer to its Ping request.
+2.030180960s Ping:Receive(0x55986e301ab0, 0x55986e159390)
+2.030180960s Ping:Receive(): recv 84 bytes
+2.030180960s Ping:Receive(): Received Echo Reply size  = 64 bytes from 10.1.1.6 id =  48879 seq =
0 TTL = 60
+2.030180960s Ping:Read64(0x55986e301ab0, 0x55986e1c69e0)
64 bytes from 10.1.1.6: icmp_seq=0 ttl=60 time=30.18 ms

----------> Now that both nodes in the CSMA network know the MAC address of the other one, there
is no need for a new ARP request.
----------> By adding the total time spend for the 2 ARP request, we have an additional delay of:
2.021614840-2.008550400 = 0.1306444s.
----------> It corresponds well to the time difference between the first round-trip delay and the
next ones.
```